| Shotton Mill Limited | **Standard Operating Procedure** | Document No: | | Prepared By: | |
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| | | Original Issues Date: | | | |
| | | Revision Date: | | Approved By: | |
| | | Revision No: | | | |

| **Department:** | Engineering / Maintenance | P.P.E. Required (Place "X" in the required boxes) | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Area:** | PM3 OCC Pulping | | | | | | | | | | | |
| **Section:** | Pulper VC-98130G | Safety Glasses | Safety Footwear | Workwear | Hard Hat | Work Gloves | Hearing Protection | Safety Harness | Mask | Hi-Vis | L.O.T.O. | Chemical Workwear |
| **Task:** | Rotor Clearance Measurement and Rotor Condition Inspection | X | X | X | | X | X | | | | X | |

| **Permit (Place "X" in required boxes)** | | | | | | | |
|---|---|---|---|---|---|---|---|
| General | X | Working At Height | | Break Containment | | High Voltage | |
| Hot Work | | Confined Space | X | Excavation | | Isolation Reference No. | Enter site isolation reference before work starts |

| **Reason for this Activity / Task / Operation** |
|---|
| This is the procedure for measuring rotor clearance and inspecting rotor condition on the Valmet Vertical Pulper VC-98130G. Rotor clearance is the distance between the rotor vane and the screen plate. An increased rotor clearance weakens the deflaking of stock, and wear to the rotor or screen plate impairs the deflaking properties of the pulper. The OEM specifies this task at 8-week intervals during shutdown. Maintaining correct clearance ensures effective stock processing, energy efficiency and prevents unplanned downtime. |
| **Safety Notes:** |
| The main safety risks/hazards identified:<br>- Entrapment hazard: Rotor and drive must be fully isolated and locked out before any person enters the pulper or accesses the rotor unit.<br>- Slips, trips and falls: Residual stock, water and pulp on flooring and inside the pulper vessel.<br>- Confined space: Entry into the pulper vessel may be required; follow site confined space procedure.<br>- Cuts and abrasions: Rotor vane edges and screen plate perforations are sharp. |

| Reason for this Activity / Task / Operation |
|---|
| - Manual handling: Rotor components and tools are heavy; use mechanical aids where available. |
| **Environmental Notes:** |
| The main environmental risks/hazards identified:<br>- Stock residue washed from the pulper must be directed to the existing drainage system, not to surface water drains.<br>- Any oil or grease spillages should be reported to Utilities, cleaned up and the area made safe. |

| Tools & Equipment Required |
|---|
| - Feeler gauge set (range including 5 mm and 10 mm blades)<br>- Straight edge (minimum length to span rotor vane to screen plate gap)<br>- Torch / inspection lamp (intrinsically safe if confined space entry required)<br>- Tape measure<br>- Calibrated depth gauge or vernier caliper<br>- Marker pen or paint marker<br>- Inspection mirror (if required for underside viewing)<br>- Digital camera for condition recording<br>- Cleaning tools (scraper, brush) for stock residue removal<br>- PPE as listed above<br>- Permit to Work documentation<br>- Previous inspection records for trend comparison |

| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
|---|---|---|---|---|---|---|
| 1. | Confirm pulper is scheduled for shutdown and stock has been emptied. | Check with Production Authorised Person (P.A.P.) and verify pulper level indication reads zero on DCS. | To ensure the pulper is empty and safe to access for maintenance. | Site photo required | | Pulper must be fully drained before proceeding. *⚠ Shutdown requirement implied by OEM PM table (8wk shutdown task) but specific DCS tag/level indicator reference not in manual — needs site confirmation.* |
| 2. | Obtain Permit to Work with necessary isolations. | Check with P.A.P. that pulper drive motor, gear unit and all auxiliary systems are isolated and locked out. | To prevent movement of rotor during inspection and ensure safety of the team. | Site photo required | | Isolation Reference No. must be recorded on permit. *⚠ Isolation/LOTO is good practice and implied by OEM safety section (p.56/p.11) but no specific isolation point tags or site PTW procedure referenced in manual.* |
| 3. | Conduct shift brief with all personnel involved. | Supervisor or Team Leader briefs team on task scope, hazards and emergency procedures. | To allocate work and highlight required safety issues. | Site photo required | | All team members must sign onto the permit. *⚠ Shift brief is sound safety practice but not explicitly required by OEM manual. Generic/common-sense step; no OEM evidence directly supports it.* |

| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
|---|---|---|---|---|---|---|
| 4. | Verify isolation is effective by attempting to start pulper drive locally. | Press start button on local control panel; confirm no response. | To prove the isolation is secure before personnel access the rotor. | Site photo required | | Drive must not respond. If it does, stop work immediately. *⚠ Isolation verification by attempted start is good practice. OEM safety section references isolation but does not specify this exact verification method. Local panel tag unconfirmed.* |
| 5. | Open pulper access hatch or manway. | Undo fasteners on the access hatch using appropriate spanners. Support hatch to prevent it falling. | To gain access to the rotor and screen plate for inspection. | Site photo required | | Confirm confined space entry controls are in place if vessel entry required. *⚠ Access hatch opening implied by OEM shutdown inspection requirement but hatch location, fastener type and size not specified in available manual evidence.* |
| 6. | Ventilate the pulper vessel if confined space entry is required. | Set up forced ventilation fan at the access opening and confirm atmosphere is tested and safe. | To ensure a safe breathing atmosphere inside the vessel. | Site photo required | | Atmosphere must be confirmed safe by gas monitor before entry. *✗ Confined space ventilation not mentioned in OEM manual. Requirement depends on site risk assessment. Customer must confirm with site safety team before including.* |

| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
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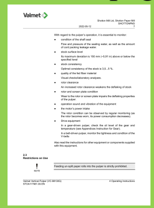
| 7. | Remove stock residue from the rotor vanes and screen plate surface. | Using scraper and brush, clean the top surfaces of the rotor vanes and the visible screen plate area. | To expose the true metal surfaces for accurate clearance measurement and condition assessment. | Site photo required | | All six rotor vanes must be cleaned sufficiently to allow measurement.<br>⚠ *Cleaning before measurement is logical and implied by OEM wear check requirement (p.323) but the manual does not explicitly describe a pre-measurement cleaning step.* |
| 8. | Visually inspect the rotor vanes for wear, cracking or damage. | Using torch and inspection mirror, examine all six rotor blades for erosion, cracks, chipping or deformation. | Wear to the rotor impairs the deflaking properties of the pulper. | Manual page 49<br> | | Record any visible wear patterns, cracks or material loss.<br>✓ *Directly supported: OEM p.49 states rotor/screen plate condition check required; p.34 confirms six-bladed rotor. Wear inspection is explicitly listed in PM table p.323 at 8wk.* |
| 9. | Photograph the rotor vane condition. | Using digital camera, take close-up photos of each vane showing wear areas. | To provide a visual record for trending and comparison with previous inspections. | Site photo required | | Include a scale reference in photos where possible.<br>⚠ *Photography for trending is good practice but not specified in OEM manual. Reasonable addition; customer should confirm it aligns with site inspection record requirements.* |

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| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
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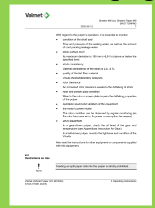
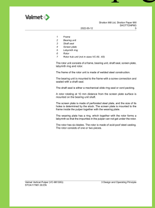
| 10. | Visually inspect the screen plate for wear, erosion or hole enlargement. | Using torch, examine the screen plate surface visible around and beneath the rotor. | Wear to the screen plate impairs deflaking and can allow oversize contaminants to pass. | Manual page 49  | | Note any areas of thinning, cracking or enlarged perforations.<br>*✓ Directly supported: OEM p.49 and p.323 explicitly require rotor and screen plate wear check at 8-week shutdown interval. Impaired deflaking consequence stated on p.49.* |
| 11. | Photograph the screen plate condition. | Using digital camera, take photos of any wear areas on the screen plate. | To provide a visual record for trending. | Site photo required | | *⚠ Screen plate photography is reasonable for trending but not OEM-specified. Key point field is blank — should state minimum photo requirement for record completeness.* |
| 12. | Inspect the labyrinth ring between rotor and wearing plate. | Visually check the labyrinth ring for wear, damage or stock build-up that could allow contaminants under the rotor. | The labyrinth ring prevents impurities from getting under the rotor. | Manual page 34  | | Report any damage or excessive clearance at the labyrinth.<br>*✓ Directly supported: OEM p.34 explicitly describes labyrinth ring function — prevents impurities getting under rotor. Inspection is consistent with OEM wear check requirement.* |

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| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
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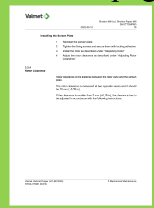
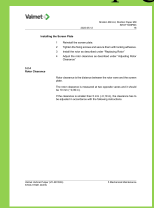
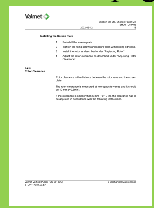
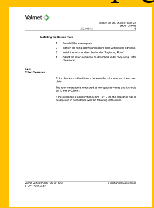
| 13. | Select the first rotor vane for clearance measurement. | Mark the first vane with paint marker for reference. Identify the diametrically opposite vane. | OEM requires measurement at two opposite vanes. | Manual page 77  | | Measurements must be taken at two opposite vanes per OEM instruction. *✓ Directly supported by OEM p.77: clearance measured at two opposite vanes. Paint marker reference method is reasonable site practice to implement this OEM requirement.* |
| 14. | Measure rotor clearance at the first vane. | Insert feeler gauge between the underside of the rotor vane and the screen plate surface. Read the gap. | To determine the current clearance and compare against the OEM specification. | Manual page 77  | | Target clearance is 10 mm. Minimum acceptable is 5 mm. *✓ Directly supported: OEM p.77 specifies 10mm target clearance, 5mm minimum trigger for adjustment. Feeler gauge method is consistent with OEM measurement description.* |
| 15. | Record the clearance reading for the first vane. | Write the measurement value on the inspection record sheet against vane position 1. | To maintain a documented history for trend analysis. | Site photo required | | *⚠ Recording is good practice and implied by OEM PM table calibration requirement (p.323) but no specific OEM inspection form or CMMS reference is defined in the manual.* |

| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
|---|---|---|---|---|---|---|
| 16. | Measure rotor clearance at the opposite vane. | Insert feeler gauge between the underside of the diametrically opposite rotor vane and the screen plate. | To check for uneven wear across the rotor. | Manual page 77  | | Target clearance is 10 mm. Minimum acceptable is 5 mm. *✓ Directly supported: OEM p.77 explicitly requires measurement at two opposite vanes. Target 10mm and minimum 5mm values are directly stated in the manual.* |
| 17. | Record the clearance reading for the opposite vane. | Write the measurement value on the inspection record sheet against vane position 2. | To maintain a documented history for trend analysis. | Site photo required | | *⚠ Recording implied by OEM calibration requirement (p.323) but specific form/CMMS not defined in manual. Consistent with step 15 — site must confirm record format.* |
| 18. | Measure clearance at remaining vane pairs if accessible. | Rotate rotor by hand (if possible) to expose additional vanes and repeat feeler gauge measurement. | To identify localised wear that may not be apparent from two measurements alone. | Manual page 77  | | Record all readings. Note if rotor cannot be rotated by hand. *⚠ Additional vane measurements are good practice but OEM only specifies two opposite vanes (p.77). Manual rotation method and tooling not confirmed — flagged assumption.* |

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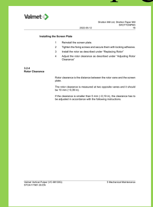
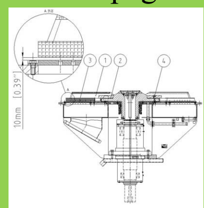
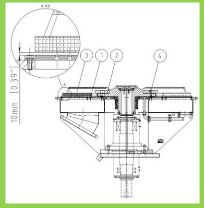
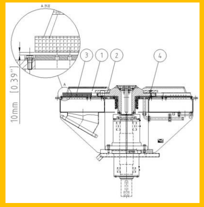
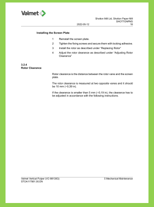
| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
|---|---|---|---|---|---|---|
| 19. | Compare measured clearances against previous inspection records. | Review the previous 8-weekly inspection data and calculate the rate of wear. | To identify accelerating wear trends that may require earlier intervention. | Site photo required | | Increasing trend toward 5 mm minimum triggers corrective action planning.<br>⚠ *Trend comparison is sound maintenance practice. OEM PM table (p.323) implies ongoing calibration records but does not specify 8-weekly comparison methodology explicitly.* |
| 20. | Determine if clearance adjustment is required. | If any measurement is less than 5 mm, clearance must be adjusted per OEM instructions. | Clearance below 5 mm risks rotor-to-screen-plate contact and damage. | Manual page 77<br> | | If clearance < 5 mm, proceed to shim adjustment. If ≥ 5 mm, no adjustment needed.<br>✓ *Directly supported: OEM p.77 explicitly states if clearance less than 5mm adjustment is required. Decision logic in step matches OEM instruction precisely.* |
| 21. | If adjustment required: remove rotor fixing screws. | Using appropriate spanner, undo the rotor fixing screws securing the rotor to the hub unit. | To allow access to the shim plates for clearance adjustment. | Manual page 79<br> | | Keep screws in a labelled container to prevent loss.<br>✓ *Directly supported: OEM p.79 describes unscrewing rotor fixing screws as step 1 of two-piece rotor clearance adjustment procedure for VC-77/-87/-98 series.* |

| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
|---|---|---|---|---|---|---|
| 22. | If adjustment required: add or remove shim plates as needed. | Add shims to reduce clearance or remove shims to increase clearance between rotor and hub unit. | Shim adjustment sets the rotor vane-to-screen-plate distance back to 10 mm. | Manual page 79  | | Target clearance after adjustment is 10 mm. *✓ Directly supported: OEM p.79 (two-piece rotor) and p.78 (one-piece) describe adding/removing shims to adjust clearance. Target 10mm confirmed on p.77.* |
| 23. | If adjustment required: reinstall rotor fixing screws and tighten. | Refit screws and tighten to the torque specified in the Tightening Torques section of the manual. | To secure the rotor in its adjusted position. | Manual page 79  | | Refer to manual Section 5 Tightening Torques for correct values. *⚠ Reinstallation of screws supported by OEM p.79 step 4. However, specific torque value not provided in available evidence — must be confirmed from manual Section 5 p.52.* |
| 24. | If adjustment required: re-measure clearance at two opposite vanes to confirm. | Repeat feeler gauge measurement at the same two opposite vanes. | To verify the adjustment has achieved the target clearance. | Manual page 77  | | Confirmed clearance must be 10 mm at both measurement points. *✓ Directly supported: OEM p.79 step 5 explicitly states check rotor clearance and adjust again if necessary. Re-measurement at two opposite vanes consistent with p.77.* |

| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
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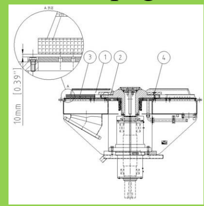
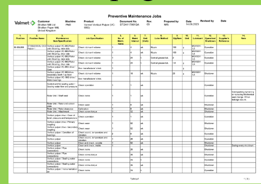
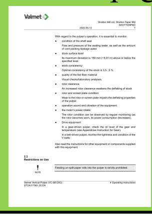
| 25. | If adjustment required: secure the rotor fixing screws. | Apply locking method as specified by OEM (e.g. locking adhesive or lock wire). | To prevent screws loosening during operation. | Manual page 79  | | Screws must be positively secured against vibration. *✓ Directly supported: OEM p.79 step 6 explicitly states secure the screws. Locking method (adhesive/lockwire) consistent with OEM screen plate instruction p.77.* |
| 26. | Check rotor attachment screw torque (annual check if due). | Using calibrated torque wrench, verify rotor unit attachment screws are at specified torque. | OEM specifies attachment screw torque check every 52 weeks. | Manual page 323  | | Only required if 52-week interval has elapsed. Record result. *✓ Directly supported: OEM PM table p.323 explicitly lists rotor unit attachment screw torque check at 52-week interval, shutdown condition. Interval logic in step is correct.* |
| 27. | Inspect shaft seal area for leaks. | Visually check around the shaft seal for signs of water or stock leakage. | Shaft seal condition is monitored weekly; opportunity inspection during shutdown. | Manual page 49  | | Report any leakage for corrective action. *✓ Directly supported: OEM PM table p.323 lists shaft seal leak check at 1-week interval. Opportunity inspection during shutdown is consistent with OEM monitoring requirement.* |

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| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
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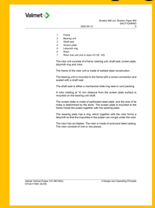
| 28. | Check for abnormal wear or scoring on the rotor hub mating surfaces. | Visually inspect the hub-to-rotor interface for fretting, corrosion or scoring. | Hub wear can affect rotor seating and clearance accuracy. | Manual page 34  | | Report any fretting or corrosion for engineering review. *⚠ Hub mating surface inspection is reasonable engineering practice but not explicitly listed as a separate OEM PM task. No direct manual evidence for this specific check.* |
| 29. | Remove all tools and equipment from inside the pulper. | Physically check inside the vessel and collect all tools, gauges and debris. | To prevent foreign objects damaging the rotor or screen plate on restart. | Site photo required | | Perform a tool count against the tool list before closing. *⚠ Tool clearance before closing is essential safety practice but not explicitly stated in OEM manual. Good practice; customer should confirm against site FOD/tool control procedure.* |
| 30. | Close and secure the pulper access hatch. | Refit hatch and tighten all fasteners evenly to the correct torque. | To seal the pulper for operation and prevent leaks. | Site photo required | | All fasteners must be fitted; no gaps or misalignment. *⚠ Hatch closure implied by OEM shutdown task completion but specific fastener torque for access hatch not provided in available manual evidence — needs site confirmation.* |
| 31. | Remove confined space ventilation equipment if used. | Switch off fan, disconnect ducting and store equipment. | To clear the area for normal operation. | Site photo required | | *⚠ Ventilation equipment removal is logical follow-on to step 6 but not OEM-specified. Only relevant if confined space entry was required — site must confirm applicability.* |

| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | Key Point |
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| 32. | Complete the inspection record with all measurements and observations. | Fill in all fields on the PM inspection form including clearance values, condition notes and photos taken. | To provide documented evidence for the CMMS and trend analysis. | Site photo required | | Record must include date, personnel, clearance values and pass/fail status. *⚠ Documentation requirement implied by OEM PM calibration entry (p.323) but specific form, CMMS system or required fields not defined in OEM manual — site must confirm.* |
| 33. | Raise any corrective work orders identified during inspection. | Enter defects into the CMMS (e.g. rotor replacement, screen plate replacement, seal repair). | To ensure follow-up actions are planned and scheduled. | Site photo required | | Priority based on proximity to 5 mm minimum clearance limit. *⚠ Raising corrective work orders is sound CMMS practice. Not explicitly in OEM manual but consistent with OEM PM table structure. CMMS system name must be confirmed locally.* |
| 34. | Return Permit to Work to the Production Authorised Person. | Sign off the permit confirming all work complete, area clear and equipment ready for service. | To formally hand back the equipment for de-isolation and restart. | Site photo required | | Permit must be signed off before isolations are removed. *⚠ PTW sign-off is correct safety practice and consistent with OEM safety section reference (p.56/p.11) but site-specific PTW procedure and P.A.P. role not defined in OEM manual.* |

| Step | Action | How? | Why? | Picture / Layout | Time (Hrs:Mins) | ⚠ Key Point |
|---|---|---|---|---|---|---|
| 35. | Confirm de-isolation and reinstatement of pulper drive. | P.A.P. removes locks and restores power to drive motor and auxiliaries per site procedure. | To return the pulper to operational readiness. | Site photo required | | Operator to confirm pulper runs normally after restart. *⚠ De-isolation and restart are implied by OEM operational requirements but specific reinstatement sequence, auxiliary systems checklist and DCS confirmation not in OEM manual.* |

**Review Summary:**
The SOP is well-structured and the core technical steps — rotor clearance measurement at two opposite vanes, 10mm target, 5mm minimum trigger, shim adjustment, screw securing and 52-week torque check — are all directly and strongly supported by OEM manual evidence. Safety and administrative steps (PTW, confined space, CMMS, photography) are reasonable but rely on site-specific procedures not covered by the OEM manual and must be reviewed and populated by the customer. The main gaps requiring customer action are: confirmation of rotor fixing screw torque values from manual Section 5, confined space entry applicability, local control panel and DCS tag references, and site inspection form/CMMS details.
**Areas for Improvement:**
• Add the specific rotor fixing screw torque value from manual Section 5 (page 52) to steps 23 and 26 — currently referenced but not populated, which will prevent the SOP being used without manual look-up.
• Confirm whether vessel entry is required for clearance measurement on this specific VC-98130G installation and update steps 5, 6, 7 and 31 accordingly — if measurement is possible from the access hatch without entry, the confined space steps may be unnecessary and should be removed or made conditional.
• Add the site-specific DCS tag or level indicator reference to step 1 and the local control panel designation to step 4 to make these steps unambiguous for the operator without needing to consult additional documents.
• Populate the blank key point field in step 11 (screen plate photography) with a minimum requirement, e.g. 'At least one photo per wear area identified; photos to be filed against the PM work order in the CMMS.'

• Add a note to step 18 confirming the approved method for rotating the rotor by hand on this installation — whether a specific jacking tool, bar through the rotor, or other method is required — as this is flagged as an unconfirmed assumption.